



Computer Science Department
COMP233-Discrete Mathematics/ Spring 2026
Section #4 - Mid #1: Chapters 2, 3, 4
Time: 60 Minutes | Total Marks: 60

Student Name:

Student ID:

Question 1#: Fill **ALL** correct answers for each branch in the table in the answer sheet.
[15 Marks]

1. A contradiction is a statement that is always:			
a) True	b) False	c) Conditional	d) Valid
2. The negation of "All dogs are loyal" is (Circle all that apply):			
a) All dogs are disloyal	b) Some dogs are disloyal	c) No dogs are loyal	d) There is a dog that is disloyal
3. Which of these is equivalent to "$\forall$ basketball players x, x is tall"? (Circle all that apply)			
a) Anyone who is tall is a basketball player	b) Every basketball player is tall	c) All people who are basketball players are tall	d) Some tall people are basketball players
4. The parity of an integer indicates whether it is:			
a) Positive or negative	b) Rational or irrational	c) Even or odd	d) Prime or composite
5. How many zeros are at the end of the number $45^8 \cdot 88^5$?			
a) 5	b) 8	c) 10	d) 0
6. An if-then statement is FALSE only when the hypothesis is ____ and the conclusion is ____:			
a) True, True	b) False, True	c) True, False	d) False, False
7. To disprove $\forall x \in D, P(x)$, we need to find:			
a) A truth set	b) A tautology	c) A counterexample	d) A critical row
8. The statement $p \rightarrow q$ is logically equivalent to: (Circle all that apply)			
a) $\sim p \vee q$	b) $p \wedge \sim q$	c) $\sim q \rightarrow \sim p$	d) $q \rightarrow p$
9. The average of any two odd integers is odd." Which of the following is true about this statement? (Circle all that apply)			
a) It is true for all odd integers	b) It is false; a counterexample is: $x = 3$ and $y = 5$	c) It is false; a counterexample is: $x = 1$ and $y = 7$	d) Both (b) and (c) are correct
10. Consider the expression $6m(2m + 10)$ where m is any integer. This expression is:			
a) Always divisible by 4	b) Divisible by 4 only if m is even	c) Divisible by 4 only if m is odd	d) Never divisible by 4



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Question #2: Basic Logic Section

[18 Marks]

A) Write the formal negation for the following:

1. $\forall$ computers c , c has a CPU.
2. $\exists x \in \mathbb{R}$ such that $x^2 = 2$.

B) Using logical equivalence laws, prove the following equivalence. Justify each step by naming the law used:

$$\sim(\sim p \vee q) \wedge (p \vee q) \equiv p$$

Question #3: Concepts Section

[10 Marks]

A) Consider the statement: "If n is a prime number and $n > 2$, then n is odd."

1. Write the Contrapositive.
2. Write the Negation.

B) Use a Truth Table to determine if the following argument form is valid or invalid. Indicate which columns represent the premises and which represent the conclusion (Flag the critical rows):

$$\begin{array}{l} p \rightarrow q \\ \sim p \\ \therefore \sim q \end{array}$$

Question #4: Proofs Section

[17 Marks]

A) Indicate whether the following statements are True or False (Justify briefly):

1. $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}$ such that $x + y = 0$.
2. $\exists y \in \mathbb{Z}$ such that $\forall x \in \mathbb{Z}, x + y = 0$.

B) Prove the following:

1. If $n = 4k + 1$ for some integer k , then 8 divides $(n^2 - 1)$.
2. Use the **quotient-remainder theorem** with $d = 3$ to prove that for any integer n , the product of two consecutive integers $n(n+1)$ has the form $3k$ or $3k+2$ for some integer k .

Please, use the answer sheet form ONLY!

Good Luck!



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Answer Sheet

Question 1#: Fill your final answers in the table in the answer sheet. [15 Marks]

Question	1	2	3	4	5	6	7	8	9	10
Answer										

Question #2: Basic Logic Section [18 Marks]

A) Formal Negation (6 Marks):

1.

2.

B) Logical Equivalence Reasons (9 Marks):

$$\sim(\sim p \vee q) \wedge (p \vee q) \equiv p$$

$$\sim(\sim p \vee q) \wedge (p \vee q) \equiv (\sim(\sim p) \vee \sim q) \wedge (p \vee q) \text{ by (_____)}$$

$$\equiv \text{_____ by (_____)}$$

$$\equiv \text{_____ by (_____)}$$

$$\equiv p \vee (q \wedge \sim q) \text{ by (the commutative law for } \wedge \text{)}$$

$$\equiv \text{_____ by (_____)}$$

$$\equiv \text{_____ by (_____)}$$



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Question #3: Concepts Section

[10 Marks]

A) Statement Variations (4 Marks):

1. **Contrapositive:**

2. **Negation:**

B) Truth Table & Validity (6 Marks):

p	q			
T	T			
T	F			
F	T			
F	F			

▪ **Result (Valid or Invalid):**

▪ **Justification (Why?):**

Question #4: Proofs Section

[17 Marks]

A) True/False (4 Marks):

1.

2.



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B) Divisibility Proof (6 Marks):

C) Consecutive Integers Proof (7 Marks):



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